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**Instructions:**

- DO NOT TURN TO THE SECOND PAGE UNTIL YOUR INSTRUCTOR HAS ALLOWED YOU TO DO SO.
- You are allowed a one cheat sheet, but no other notes or resources. You are NOT allowed to use any other external resource like cellphones, lecture notes, books or talk to any of your peers.
- Show your work. Full points are only given for correct answers *with* adequate justification. A correct final answer with missing or substantially incorrect justification will not merit full points on a problem.
- If you need an extra page to complete your computations for a problem you will find another page at the end of this midterm. Indicate clearly if you used this extra space while solving a problem, for example leaving a message on the page of Problem 3: *the rest of my solution to Problem 3 is in the extra-page.*
- Good luck! Read the entire midterm first and start with those points that make you feel more confident.

**Problem 1. (5 pts)** Let  $(G, \star, e)$  be a group. Consider the following property:

(P)  $G$  is not abelian but there is a proper subgroup  $H \leq G$  that is abelian.

1. (1 pt) State the definition of a structural property.
2. (4 pts) Show that the property  $P$  is a structural property.

**Solution:**

The property of being abelian is invariant under automorphisms (this is one of the examples that we did in class), so if  $G$  is not abelian and  $G' \cong G$  then  $G'$  is also not abelian (if it were so will be  $G$ ). Thus the property of not being abelian is a structural property.

For the second part, assume that  $G$  has an abelian subgroup  $H$  that is abelian. Let  $f : G \rightarrow G'$  be the isomorphism, we aim to show that  $f(H)$  is an abelian subgroup of  $G'$ .

*$f(H)$  is a subgroup of  $G'$ :*

It contains the identity, since  $e_{G'} = f(e_G)$ . It is closed under the operation, since given  $g_1, g_2 \in f(H)$  there are  $h_1, h_2 \in H$  such that  $f(h_1) = g_1$  and  $f(h_2) = g_2$ . Then  $f(h_1 \cdot h_2) = f(h_1) \cdot f(h_2) = g_1 \cdot g_2$ , thus  $g_1 \cdot g_2 \in f(H)$ , because as  $H$  is a subgroup  $h_1 \cdot h_2 \in H$ .

Lastly we need to check that it is closed under inverses, let  $g \in f(H)$ , then there is  $h \in H$  such that  $f(h) = g$ . Since  $H$  is subgroup  $h^{-1} \in H$ , then  $f(h) \cdot f(h^{-1}) = f(h \cdot h^{-1}) = f(e_G) = e_{G'}$ , thus the inverse of  $g = f(h)$  is  $f(h^{-1}) \in f(H)$ , as required.

Lastly, we need to show that  $f(H)$  is abelian. So let  $g_1, g_2 \in f(H)$  we want to show that  $g_1 \cdot g_2 = g_2 \cdot g_1$ . Since  $f$  is surjective, there are  $h_1, h_2 \in H$  such that  $f(h_1) = g_1$  and  $f(h_2) = g_2$ . Since  $H$  is abelian  $h_1 \cdot h_2 = h_2 \cdot h_1$ , applying  $f$  in both sides  $f(h_1 \cdot h_2) = f(h_2 \cdot h_1)$ , since  $f$  is an homomorphism  $f(h_1) \cdot f(h_2) = f(h_2) \cdot f(h_1)$ . Consequently,  $g_1 \cdot g_2 = g_2 \cdot g_1$  thus  $f(H)$  is abelian.

**Problem 2. (7 pts)** Let  $(G, \cdot, e)$  be a group.

**Recall:** an element  $g \in G$  is said to be of order  $k$  if  $k$  is the smallest positive integer such that  $g^k = e$ . Let  $g \in G$  be an element of order  $k$ .

1. (4 pts) If  $\phi : G \mapsto H$  is an homomorphism prove that the order of  $\phi(g)$  divides  $k$ .

**Solution:**

Assume  $g^k = e_G$ , and that  $k$  is minimal i.e. it is the order of  $g$  in  $G$ . Since  $\phi$  is an homomorphism

$$\phi(g)^k = \phi(e_G) = e_H \tag{1}$$

Let  $n$  be the order of  $\phi(g)$ , i.e.  $n$  is the smallest natural positive number such that  $\phi(g)^n = e_H$ . By the equation above  $n \leq k$ . We must argue that  $n$  divides  $k$ .

By the Euclid Algorithm, there are  $q, r \in \mathbb{Z}$  such that  $k = nq + r$  where  $0 \leq r < n$ .

Then  $e_H = \phi(g)^k = \phi(g)^{nq+r} = (\phi(g)^n)^q \cdot \phi(g)^r = e_H \cdot \phi(g)^r = \phi(g)^r$ .

By minimality of  $n$ , then  $r = 0$ . In particular,  $k = nq$ , so the order of  $\phi(g)$  in  $H$  divides the order of  $g$  in  $G$ .

2. (3 pts) If  $\phi : G \mapsto H$  is an isomorphism prove that the order of  $\phi(g)$  is equal to  $k$ .  
Let  $n$  be the order of  $\phi(g)$  in  $H$ , by the previous questions  $n$  divides  $k$ . Since  $\phi$  is an isomorphism, then  $\phi^{-1}$  is an isomorphism from  $H$  to  $G$ , then by the previous question  $k$  divides  $n$ . Then  $k = n$ .

**Problem 3. (12 pts)**

Let  $(G, \star, e)$  be a group and  $H \leq G$  be a subgroup. Given  $g \in G$ , let

$$gHg^{-1} = \{ghg^{-1} \mid h \in H\}$$

1. (1 pts) State the definition of a subgroup of  $G$ .

**Solution:**

A subset  $H \subseteq G$  is a subgroup of  $G$  if it satisfies the following 3 conditions:

- $e_G \in H$ ,
- it is closed under operation: if  $h_1, h_2 \in H$  then  $h_1h_2 \in H$ ,
- it is closed under inverses: if  $h \in H$  then  $h^{-1} \in H$ .

2. (5 pts) Prove that for each  $g \in G$ , the group  $gHg^{-1}$  is a subgroup of  $G$ .

**Solution:**

Note that  $e_G = ge_Gg^{-1} \in gHg^{-1}$  since  $e_G \in H$  because it is a subgroup.

Let  $z_1, z_2 \in gHg^{-1}$ , we want to show that  $z_1 \cdot z_2 \in gHg^{-1}$ . There are  $h_1, h_2 \in H$  such that  $gh_1g^{-1} = z_1$  and  $gh_2g^{-1} = z_2$ . Then  $z_1z_2 = gh_1g^{-1}gh_2g^{-1} = gh_1h_2g^{-1}$ . Since  $H$  is a subgroup it is closed under the operation, then  $h_1h_2 \in H$ , so  $gh_1h_2g^{-1} \in gHg^{-1}$ .

We just need to verify for inverses, since  $H$  is a subgroup for every  $h \in H$  we have  $h^{-1} \in H$ . The inverse of  $ghg^{-1}$  is  $gh^{-1}g^{-1}$  since  $ghg^{-1}gh^{-1}g^{-1} = e_G$ . By uniqueness of inverses,  $(ghg^{-1})^{-1} = gh^{-1}g^{-1} \in gHg^{-1}$ , as required.

3. (1 point) Let  $S$  be a set and  $(G, \cdot, e)$  be a group. What is an action of  $G$  on  $S$ ?

**Solution:**

Let  $\mu : G \times S \rightarrow S$  be a map, we say that it is an action if the following conditions are satisfied:

- (a) for every  $g, h \in G$  and  $s \in S$  we have  $\mu(g \cdot h, s) = \mu(g, \mu(h, s))$ .
- (b) for every  $s \in S$ ,  $\mu(e, s) = s$ .

4. (5pts) Let  $Sub(G)$  denote the set of all subgroups of  $G$ , prove that this construction defines an action of  $G$  on  $Sub(G)$ , i.e. the map

$$\begin{aligned}\mu : G \times Sub(G) &\mapsto Sub(G) \\ (g, H) &\rightarrow gHg^{-1}\end{aligned}$$

is an action of  $G$  into  $Sub(G)$ .

**Solution:**

Let  $H \in Sub(G)$  and let  $g_1, g_2 \in G$ , we want to show that  $\mu(g_1 \cdot g_2, S) = \mu(g_1, \mu(g_2, S))$ .

On one hand we get  $\mu(g_1 \cdot g_2, S) = (g_1 \cdot g_2)S(g_1 \cdot g_2)^{-1} = (g_1 \cdot g_2)S(g_2^{-1}g_1^{-1}) = g_1(g_2Sg_2^{-1})g_1^{-1} = g_1\mu(g_2, S)g_1^{-1} = \mu(g_1, \mu(g_2, S))$ , as required.

Let  $e_G$  be the identity element of  $G$ , and fix  $S \in Sub(G)$  then  $\mu(e_G, S) = e_GSe_G^{-1} = S$ .

We conclude that  $\mu$  is a left action.



**Problem 4. (11 pts)** Let  $n, m \in \mathbb{N}$  and consider the binary operation

$$\star : (\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}) \mapsto (\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z})$$

$$(([a]_{\text{mod } n}, [b]_{\text{mod } m}), ([w]_{\text{mod } n}, [x]_{\text{mod } m})) \rightarrow ([a+w]_{\text{mod } n}, [b+x]_{\text{mod } m})$$

and let

$$\phi : \mathbb{Z} \mapsto \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$$

$$z \rightarrow ([z]_{\text{mod } n}, [z]_{\text{mod } m}).$$

1. (3 pts) Show that  $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \star)$  is a group.

**Solution:**

We must verify 3 conditions:

- (a) the operation  $\star$  is associative: let  $\bar{x} = ([x_1]_{\text{mod } n}, [x_2]_{\text{mod } m})$ ,  $\bar{y} = ([y_1]_{\text{mod } n}, [y_2]_{\text{mod } m})$  and  $\bar{z} = ([z_1]_{\text{mod } n}, [z_2]_{\text{mod } m})$ .

We want to show that  $\bar{x} \star (\bar{y} \star \bar{z}) = (\bar{x} \star \bar{y}) \star \bar{z}$ .

$$\bar{x} \star (\bar{y} \star \bar{z}) = \bar{x} \star ([y_1+z_1]_{\text{mod } n}, [y_2+z_2]_{\text{mod } m}) = ([x_1+(y_1+z_1)]_{\text{mod } n}, [x_2+(y_2+z_2)]_{\text{mod } m}) =$$

$$[y_2+z_2]_{\text{mod } m}) = ([x_1+y_1+z_1]_{\text{mod } n}, [(x_2+y_2)+z_2]_{\text{mod } m}) = ([x_1+y_1]_{\text{mod } n}, [(x_2+y_2)]_{\text{mod } m}) \star \bar{z} = (\bar{x} \star \bar{y}) \star \bar{z}$$

- (b) the existence of an identity: let  $e = ([0]_{\text{mod } n}, [0]_{\text{mod } m})$ . Let  $\bar{x} = ([x_1]_{\text{mod } n}, [x_2]_{\text{mod } m}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ , then  $\bar{x} \star e = ([x_1+0]_{\text{mod } n}, [x_2+0]_{\text{mod } m}) = ([x_1]_{\text{mod } n}, [x_2]_{\text{mod } m}) = \bar{x}$ . Likewise,  $e \star \bar{x} = \bar{x}$ .

- (c) the existence of inverses:

Let  $\bar{x} = ([x_1]_{\text{mod } n}, [x_2]_{\text{mod } m}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ , and let  $\bar{z} = ([n-x_1]_{\text{mod } n}, [m-x_2]_{\text{mod } m})$ . Then  $\bar{x} \star \bar{z} = ([x_1+n-x_1]_{\text{mod } n}, [x_2+m-x_2]_{\text{mod } m}) = ([n]_{\text{mod } n}, [m]_{\text{mod } m}) = ([0]_{\text{mod } n}, [0]_{\text{mod } m}) = e$ .

Likewise,  $\bar{z} \star \bar{x} = e$ .

2. (3 pts) Show that  $\phi$  is an homomorphism between  $(\mathbb{Z}, +)$  and  $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \star)$ .

**Solution:**

Let  $x, y \in \mathbb{Z}$ , we want to show that  $\phi(x + y) = \phi(x) \star \phi(y)$ .

By definition of  $\phi$  and the operation of  $\star$  we have:

$$\phi(x+y) = ([x+y]_{mod n}, [x+y]_{mod m}) = ([x]_{mod n}, [x]_{mod m}) \star ([y]_{mod n}, [y]_{mod m}) = \phi(x) \star \phi(y).$$

3. (2 pts) Show that  $\phi$  induces an equivalence relation on  $\mathbb{Z}$ , defined by

$$E_\phi = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid \phi(x) = \phi(y)\}.$$

**Solution:** This you proved in a more general form in homework 1.

We must verify three conditions:

- (a) *reflexive:* Given  $x \in \mathbb{Z}$ , then  $\phi(x) = \phi(x)$  so  $(x, x) \in E_\phi$ .
- (b) *symmetry:* Given  $x, y \in \mathbb{Z}$  assume that  $(x, y) \in E_\phi$ . Then  $\phi(x) = \phi(y)$ , but then  $\phi(y) = \phi(x)$  thus  $(y, x) \in E_\phi$ .
- (c) *transitivity:* Given  $x, y, z \in \mathbb{Z}$  assume that  $(x, y) \in E_\phi$  and  $(y, z) \in E_\phi$ . Then  $\phi(x) = \phi(y)$  and  $\phi(y) = \phi(z)$ , by transitivity of the equality we have  $\phi(x) = \phi(z)$  thus  $(x, z) \in E_\phi$ .

4. (2 pt) Describe all the subgroups of  $(\mathbb{Z}, +, 0)$ . Could you characterize those subgroups  $H \leq (\mathbb{Z}, +, 0)$  such that  $\phi(H)$  is not trivial? i.e.  $\phi(H) \neq \{([0]_n, [0]_m)\}$ .

**Solution:**

The subgroups of  $(\mathbb{Z}, +, 0)$  are of the form  $k\mathbb{Z}$ . (by the theorem seen in class).

Claim:  $\phi(k\mathbb{Z})$  is trivial if and only if  $k$  is divisible by both  $n$  and  $m$ .

For the left to right direction. We show that if  $k$  is divisible by  $n$  and  $m$ , then  $\phi(H)$  is trivial. Assume that  $k = nq_1$  and  $k = mq_2$  for some  $q_1, q_2 \in \mathbb{Z}$  then for any  $z \in \mathbb{Z}$ ,  $\phi(kz) = ([nq_1z]_{\text{mod } n}, [mq_2z]_{\text{mod } m}) = ([0]_{\text{mod } n}, [0]_{\text{mod } m})$ .

For the converse, assume that  $\phi(k\mathbb{Z}) = ([0]_n, [0]_m)$ . In particular,  $\phi(k) = ([k]_{\text{mod } n}, [k]_{\text{mod } m}) = ([0]_{\text{mod } n}, [0]_{\text{mod } m})$ . Consequently,  $k \equiv_n 0$  and  $k \equiv_m 0$ , so  $k$  is divisible by both  $n$  and  $m$ .